

Cadenn Clower

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Professional Summary:

Mechanical engineering student with two Flex-N-Gate internships in design, structural analysis, and manufacturing support. Seeking internships in design, analysis, or manufacturing. Experience includes CATIA V5, 3DEXPERIENCE, ANSYS, GD&T, packaging studies, prototype testing, fixture development, and design reviews.

Education:

Bachelor of Science in Mechanical Engineering | Expected May 2028

Oakland University | Rochester, MI | GPA: 3.63

Engineering Experience:

CAD Engineering Intern

May 2026 - Present

Flex-N-Gate | Warren, MI

- Designed a Nissan Pathfinder bracket in CATIA V5 GSD and Part Design for an approximately 85 mm mounting shift; added locating/attachment features and resolved interference with revised customer fascia.
- Evaluated F-250 headlamp packaging alternatives; identified an approximately 14.5 mm minimum width increase while maintaining 4 mm attachment clearances, and prepared comparative CAD sections for program review.
- Developed a common camera retainer for five F-250 variant groups, now in final preproduction; evaluated camera and camera-delete configurations, fastener concepts, and bezel interfaces to improve part commonality.
- Developed Class A surfaces for multiple camera-bezel designs in CATIA V5; performed draft analysis on three configurations, evaluating wall thickness and radii while checking mold-release feasibility.
- Designed fixture blocks and a bolt-adapter bushing to reuse an existing fastener; defined seal-rework fixture dimensions, referencing original notches, ribs, and molded features for cutout position and depth.
- Redesigned an active-air-dam guide with three ribs and resolved interference between the guide and its cover.
- Developed five hood/fender cross sections for sheet molding compound (SMC) with thickness, adhesive-width, radius, and draft requirements; revised reinforcement and mounts as customer studio surfaces changed.

Tooling Engineer Intern

May 2025 - Aug 2025

Flex-N-Gate | Warren, MI

- Designed and optimized an F-150 headlamp attachment using ANSYS FEA; reduced calculated peak stress from 34 to 17.1 MPa, a 16.9 MPa reduction (approximately 50%), and confirmed stress below material yield limits.
- Designed and revised stamped-steel components in CATIA V5 for Ford, GM, and Mercedes programs; applied DFM while evaluating material properties, dimensions, tooling constraints, and production feasibility.
- Collaborated with engineering, tooling, and production teams to investigate manufacturing issues, perform root-cause analysis, and develop corrective actions supporting quality, reliability, safety, and process efficiency.

Engineering Projects:

CAD Drill Kit Design Project: Designed and assembled a multi-component drill kit in CATIA V5 using 3D parametric modeling, material selection, dimensional control, assembly constraints, and engineering drawing standards.

Additional Projects: Regenerative Steam Power Plant Analysis; R-134a Thermodynamic System Analysis; Manufacturing Process Simulation & Optimization; Arduino General-Purpose Input/Output (GPIO), Pulse-Width Modulation (PWM) LED Control & Binary Counter Programming.

Technical Skills:

CAD & Analysis: CATIA V5, 3DEXPERIENCE, SolidWorks, AutoCAD, ANSYS, MATLAB, Spinfire

Design & Manufacturing: Class A surfacing, Generative Shape Design (GSD), Part Design, surface/solid modeling, assembly development, geometric dimensioning and tolerancing (GD&T), design for manufacturability (DFM), draft analysis, fixture design

Engineering Methods: Finite element analysis (FEA), structural analysis, design verification, prototype testing, tolerance assessment, root-cause analysis, thermodynamics, fluid mechanics, heat transfer

Programming & Productivity: C++, Verilog, Microsoft Office

Activities & Achievements:

Youth Volunteer Leader - Stoney Church | Academic All-American (2024) | Dean's List - All Semesters